

PFMT — System Structure

Personal Finance Money Tracker · ServiceNow scope `x_887486_0` · every table, field and relationship

Core — the money Supporting Infrastructure PK primary key FK reference red sensitive — redacted in backups

x_887486_0_user_profile1	
sys_id	PK
username	String
password_hash	String
display_name	String
email	String
currency_preference	String
language_preference	String
ai_api_key	String
monthly_income_target	Decimal
avatar_color	String
last_login	DateTime
Owns everything below. Every REST endpoint resolves the caller's token to one profile and filters on it — which is why the API can never return another user's data.	

x_887486_0_transactionn	
sys_id	PK
user_profile	→ user_profile
account	→ account
category	→ category
transaction_type	Choice
amount	Decimal
currency	String
transaction_date	Date
description	String
notes	String
transfer_group	String
is_recurring	Boolean
recurring_frequency	Choice
next_run_date	Date
state	Choice
sys_created_on	DateTime
transaction_type : expense · income · transfer · asset · payback. transfer_group is the self-join that pairs two rows into one movement — <code>tg_ a</code> transfer, <code>ag_</code> an asset purchase.	

x_887486_0_accountn	
sys_id	PK
user_profile	→ user_profile
account_name	String
account_type	Choice
institution_name	String
current_balance	Decimal
currency	String
The account's currency is authoritative: a transaction whose currency differs from its account moves no money, so the form binds one to the other.	

x_887486_0_budgetn	
sys_id	PK
user_profile	→ user_profile
category	→ category
budget_amount	Decimal
spent_amount	Decimal
remaining_amount	Decimal
currency	String
alert_threshold	Integer
period_start	Date
period_end	Date
Keyed by <i>category</i> <i>and</i> <i>currency</i> — an SGD and an MYR budget for the same category are separate rows and never combine.	

x_887486_0_savings_goaln	
sys_id	PK
user_profile	→ user_profile
account	→ account
goal_name	String
target_amount	Decimal
current_amount	Decimal
monthly_contribution	Decimal
target_date	Date
currency	String
goal_icon	String
goal_status	Choice
remarks	String
An optional <code>account</code> link makes the goal track that account's live balance instead of a manually kept figure.	

x_887486_0_categoryn	
sys_id	PK
category_name	String
category_type	Choice
icon_emoji	String
color_hex	String
is_system_default	Boolean
Shared reference data, not owned by a user. Transactions and budgets both point at it.	

x_887486_0_sessionn	
sys_id	PK
user_profile	→ user_profile
token	String
expires_at	DateTime
device_hint	String
Live auth tokens, rebuilt on every login. Deliberately excluded from backups — archiving it would store credentials for no restore value.	

x_887486_0_backupn	
sys_id	PK
user_profile	→ user_profile
run_on	DateTime
record_count	Integer
summary	String
status	String
Optional. Holds dated copies of the other tables as attachments. Excluded from backups so they cannot nest.	

Relationships

FROM	TO	CARD.	MEANING
account.user_profile	user_profile	n : 1	Each account belongs to one user
transaction.user_profile	user_profile	n : 1	Owner — the filter every API call applies
transaction.account	account	n : 1	Which account the money moved in or out of
transaction.category	category	n : 1	What it was for
transaction.transfer_group	transaction	1 : 1	Self-join. Two rows sharing this value are one movement
budget.user_profile	user_profile	n : 1	Owner
budget.category	category	n : 1	What is being capped
savings_goal.user_profile	user_profile	n : 1	Owner
savings_goal.account	account	n : 1	Optional — goal tracks that account's balance
session.user_profile	user_profile	n : 1	One row per signed-in device

Rules the structure encodes

- **No exchange rate exists anywhere.** Currency is a plain string on each row; amounts in different currencies are never added. Every total is computed per currency.
- **A movement between accounts is two rows, not one.** A transfer writes an `expense` leg and an `income` leg sharing a `transfer_group`, so both balances move through the ordinary logic. Cross-currency legs keep their own amounts — S\$1,000 out and RM3,173.30 in — with no rate invented between them.
- **The group prefix records what kind of pair it is:** `tg_` a transfer, `ag_` an asset purchase. Reading the prefix rather than the peer's type means the pair survives even if a type fails to round-trip.
- **Neither leg is earning or spending.** Analytics excludes paired rows from income and expense; a cross-currency pair is reported separately as exchanged in / out.
- **A `payback` reduces spending rather than adding income** — money returned by someone else was never earnings.
- **Balance is derived, not stored as truth.** `current_balance` is the opening figure; the live balance is that plus income and assets, less expenses, counting only rows whose currency matches the account's.